

Notes on Bordism and Floer Field Theories

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Preface

These notes accompany the 2026 Fall reading seminar on bordism and Floer field theories. The central question is how geometric cutting and gluing give rise to functorial, Floer-theoretic, and homotopical structures. We begin with classical bordism and the Pontryagin–Thom construction, pass to topological field theory and cobordism categories, and then study their Floer-theoretic and spectral refinements.

Throughout the note, we will use the following notations and conventions:

Notation & Convention. In this note, we will use the following conventions of notation:

1. Any manifold without further comments is smooth, Hausdorff, second-countable, and paracompact.
2. A closed manifold refers to a compact manifold without boundary; an open manifold refers to a non-compact manifold without boundary.
3. Any vector space is given to be real but may not be finite-dimensional.
4. SES stands for short exact sequence; LES stands for long exact sequence.
5. Given the space Y , we denote the one-point compactification as $Y^+ := Y \cup \{\infty\}$; and we denote the space adjoining a disjoint base point as $Y_+ := Y \sqcup \{\text{pt}\}$.
6. The trivial bundle over Y is denoted as $\varepsilon_Y^r := Y \times \mathbb{R}^r$.
7. The determinant line for finite-dimensional vector space V is defined by $\text{Det}(V) = \Lambda^{\dim V} V$.
8. For oriented manifolds with boundary, the induced boundary orientation uses the outward normal first convention, i.e., $(v_{\text{out}}, v_1, \dots, v_{n-1})$ positively oriented in $T_p X$ for $p \in \partial X$ if and only if $(v_1, \dots, v_{n-1})$ is positively oriented in $T_p \partial X$. More precisely, for any short exact sequence of vector bundles

$$0 \longrightarrow E_0 \longrightarrow E \longrightarrow E_1 \longrightarrow 0$$

we use the quotient first notion of determinant line bundles $\text{Det}(E) \cong \text{Det}(E_1) \otimes \text{Det}(E_0)$.

9. Follows the previous convention of orientations, the cylinder $W := I \times X$ over orientable manifold X has orientation given by the volume form $dt \wedge \text{vol}_X$. Thus, the orientation of its boundary is given by $\partial W \cong (-X) \sqcup X$, where $-X$ denotes X with the orientation reversed.
10. For the ordered direct sum of vector spaces (or Whitney sum of vector bundles) $V \oplus W$, the orientation of the direct sum is given by the determinant line isomorphism

$$\text{Det}(V \oplus W) \cong \text{Det}(V) \otimes \text{Det}(W).$$

Part I

Pontryagin-Thom Construction and Stable Homotopy

I.1 Preliminary: Regular Value and Bordisms

In this section, we will establish the minimal theory of differential topology to defined bordisms and prove the unstable version of Pontryagin-Thom construction in the following passages.

We may start from a review of the theoretical-minimum of differential topology that we need in the following sections. For a more complete and systematic review of differential topology, one can see ?? or Milnor's book [2]. We will only introduce the part of differential topology that does not includes in the literature above.

Orientation. We may start with an alternative language of orientation via graded determinant line bundle. The orientation on smooth manifold is basically fiberwisely a vector space orientation defined on its tangent bundle, so we may start with the determinant line of vector spaces:

Definition I.1.1 (Graded Determinant Line). Let V be a finite-dimensional vector space, its graded determinant line is defined by

$$\text{Det}(V) := \Lambda^{\dim V}(V), \quad |\text{Det}(V)| = \dim V \pmod{2}$$

where $|\text{Det}(V)|$ is called the parity of the graded line.

It is not hard to prove, by the functoriality of exterior algebra, that this construction is functorial. The symmetry isomorphism of the determinant line has the Koszul sign convention:

$$\tau_{V,W} : \text{Det}(V) \otimes \text{Det}(W) \xrightarrow{\cong} \text{Det}(W) \otimes \text{Det}(V), \quad \alpha \otimes \beta \mapsto (-1)^{mn} \beta \otimes \alpha.$$

More precisely, from the property of exterior algebra, one have

$$\Lambda^\bullet(V \oplus W) \cong \Lambda^\bullet(V) \hat{\otimes} \Lambda^\bullet(W) := \bigoplus_{r=0}^{n+m} \Lambda^r(V) \otimes \Lambda^{r-k}(W).$$

where $n = \dim V$ and $m = \dim W$. Then, for top-degree isomorphism $\mu_{V,W} : \Lambda^n(V) \otimes \Lambda^m(W) \rightarrow \Lambda^{n+m}(V \oplus W)$ and $\sigma_{V,W} : V \oplus W \rightarrow W \oplus V$, there is the following commutative square of functoriality:

$$\begin{array}{ccc} \text{Det}(V) \otimes \text{Det}(W) & \xrightarrow{\mu_{V,W}} & \text{Det}(V \oplus W) \\ \tau_{V,W} \downarrow & & \downarrow \text{Det}(\sigma_{V,W}) \\ \text{Det}(W) \otimes \text{Det}(V) & \xrightarrow{\mu_{W,V}} & \text{Det}(W \oplus V) \end{array}$$

It is not hard to show that the determinant line of a vector space is indeed a line, i.e., a 1-dimensional space, which allows us to give the following definition:

Definition I.1.2 (Orientation of Vector Space). An orientation of V is a connected component in $\text{Det}(V) \setminus \{0\}$, i.e., a ray in $\text{Det}(V)$. We write the collection of orientations of V as $\mathfrak{o}(V)$.

A standard property of orientation is the following proposition:

Proposition I.1.3. Any vector space V has exactly two orientations. More precisely, the natural group action $\text{GL}(V) \curvearrowright \text{Det}(V)$ make the orientation $\mathfrak{o}(V) = \{+, -\}$ to be a $\mathbb{Z}/2$ -torsor.

As we mentioned previously, by our convention, for a SES in $\mathbf{Vect}_{\mathbb{F}}$

$$0 \longrightarrow K \longrightarrow V \longrightarrow Q \longrightarrow 0$$

the induced orientation is given by the canonical isomorphism $\mathrm{Det}(Q) \otimes \mathrm{Det}(K) \cong \mathrm{Det}(V)$, we will also write $\mathfrak{o}(V) = \mathfrak{o}(Q) \wedge \mathfrak{o}(K)$ for this isomorphism.

Transversality, Normal Bundle, and Framing. In the study of embedded submanifolds, one often cares about the intersections of submanifolds. However, certain types of "intersections" are known to be "bad", which may be unstable under perturbation or cause certain degeneracy. We only want to study the well-behaved intersections, which leads to the condition of transversality:

Definition I.1.4 (Transversality). Let M, N be smooth manifolds, $f : M \rightarrow N$ be a smooth map, and $Z \hookrightarrow N$ be an embedded submanifold. Then f is transverse to Z , denoted as $f \pitchfork Z$ if

$$(df)_x T_x M + T_{f(x)} Z = T_{f(x)} N, \quad \forall x \in f^{-1}(Z).$$

Given an embedding $i : M \hookrightarrow X$, the induced inclusion leads to the SES of vector bundles

$$0 \longrightarrow TM \xrightarrow{di} i^*TX \longrightarrow \nu_i := i^*TX/TM \longrightarrow 0$$

which defines the normal bundle.

Definition I.1.5 (Normal Bundle). The normal bundle of embedding $i : M \hookrightarrow X$ is the vector bundle $\nu_i := i^*TX/TM$ defined in the SES above.

Framed Bordisms.

I.2 Unstable Pontryagin-Thom Construction and the Stabilization

In this section, we will begin with the classical theory of bordism, whose foundations were laid by Pontryagin and Thom in the 1940s and 1950s [3, 4]. The Pontryagin–Thom construction translates bordism data into homotopy classes of maps. A regular fiber of a map to a sphere is naturally framed, while a framed submanifold determines a collapse map in the opposite direction. After stabilization, this correspondence identifies bordism groups with homotopy groups of Thom spectra [3, 4] to determine the unoriented bordism ring [4].

I.3 More Stable Homotopy Theory

Recall that in the previous section, by considering stable framing of bundles, we are able to get an intrinsic framed cobordism group in the stable case, with an isomorphism

$$\Omega_n^{\mathrm{fr}} \cong \operatorname{colim}_{k \in \mathbb{N}} \mathcal{F}_{m,k} \cong \operatorname{colim}_{k \in \mathbb{N}} \pi_{n+k}(S^k).$$

The goal of this section is to construct an object, namely the sequential spectrum $\mathbb{S}$, such that its homotopy group is naturally the stable homotopy group $\pi_n \mathbb{S} \cong \pi_n^s := \operatorname{colim}_{k \in \mathbb{N}} \pi_{n+k}(S^k)$, and then show that each sequential spectrum can be replaced by a better object called an Ω -spectrum with compatible delooping, and in this way, one can safely define derived functors in the context of stable homotopy theory.

The following conventions are used in any discussion of stable homotopy without further remark in the following passages:

1. The reduced suspension of a pointed space X is $\Sigma X := S^1 \wedge X$, and the unreduced suspension of X is $SX := (X \times I)/(X \times \partial I)$.

2. In general, space refers to based compactly generated weak Hausdorff spaces; in the case that applies to mapping telescopes, we further consider the spaces that are well-based, i.e., the basepoint inclusion is a Hurewicz cofibration.

This allows us to consider the following fact as a basic known result throughout this section:

1. $\text{Map}_*(X, Y)$ is a pointed space with basepoint being the constant map to the basepoint of Y .
2. For the loop space, $\Omega X := \text{Map}_*(S^1, X)$, the suspension-loop adjunction state that there is natural homeomorphism $\text{Map}_*(\Sigma X, Y) \cong \text{Map}_*(X, \Omega Y)$.
3. The suspension of the k -sphere satisfies $\Sigma S^k \cong S^{k+1}$.
4. Any weak equivalence is required to be bijective on π_0 , and isomorphism on π_k for $k \geq 1$. For CW complexes, we use the Whitehead theorem without proving it, which claims that all weak equivalences are homotopy equivalences.

We start with the most important definition of this section, which defines the central object we study in stable homotopy theory:

Definition I.3.1 (Sequential Spectrum). A sequential spectrum is a series of pointed spaces $E = (E_0, E_1, E_2, \dots)$ together with the structure map on each space

$$\sigma_k : \Sigma E_k \longrightarrow E_{k+1}.$$

Or equivalently, by suspension-loop adjunction

$$\alpha_k : E_k \longrightarrow \Omega E_{k+1},$$

such that $\alpha_k(x)(s) = \sigma_k([s, x])$ for any $s \in S^1$ and $x \in E_k$.

And naturally, we can define the map between spectra E and F :

Definition I.3.2 (Map between Spectra). A map of sequential spectra $f : E \rightarrow F$ is a degreewise continuous map which is compatible with structure maps, i.e. the following diagrams commute:

$$\begin{array}{ccc} \Sigma E_k & \xrightarrow{\sigma_k^E} & E_{k+1} \\ \Sigma f_k \downarrow & & \downarrow f_{k+1} \\ \Sigma F_k & \xrightarrow{\sigma_k^F} & F_{k+1} \end{array} \quad \begin{array}{ccc} E_k & \xrightarrow{\alpha_k^E} & \Omega E_{k+1} \\ f_k \downarrow & & \downarrow \Omega f_{k+1} \\ F_k & \xrightarrow{\alpha_k^F} & \Omega F_{k+1} \end{array}$$

In this way, we can define the ordinary 1-category of spectra $\mathbf{Sp}_{\text{seq}}$, with objects being sequential spectra, and the morphisms are maps between spectra. The most important example, and perhaps the motivating example of the sequential spectra, is the sphere spectra:

Example (Sphere Spectra). Defined the spectra $\mathbb{S} := (S^0, S^1, S^2, \dots)$, with the structure map be the canonical homeomorphisms $\sigma_k : \Sigma S^k \xrightarrow{\cong} S^{k+1}$.

This example leads to a functor $\mathbf{Top}_* \rightarrow \mathbf{Sp}_{\text{seq}}$.

Proposition I.3.3 (Suspension Spectra). The functor $\Sigma^\infty : \mathbf{Top}_* \hookrightarrow \mathbf{Sp}_{\text{seq}}$ defined by sending pointed space X to the spectrum $\Sigma^\infty X := (X, \Sigma X, \Sigma^2 X, \dots)$ and take f to $(\Sigma^\infty f)_k := \Sigma^k f$ is a fully faithful embedding.

Proof. The isomorphism $\Sigma S^k \cong S^{k+1}$ leads to the canonical identification $\Sigma(\Sigma^n X) \cong \Sigma^{n+1} X$ defined inductively for $n \geq 0$, and thus $\Sigma^\infty X$ is indeed a spectrum. To check that this functor is full and faithful, given a pointed map $f : X \rightarrow Y$, note that it naturally extends to each entry of the spectra by defining $(\Sigma^\infty f)_k := \Sigma^k f : \Sigma^k X \rightarrow \Sigma^k Y$. Then, for $f \neq k$, the series of map between spectra

$$\Sigma^\infty f = (f, \Sigma f, \Sigma^2 f, \dots) \neq (g, \Sigma g, \Sigma^2 g, \dots) = \Sigma^\infty g$$

which shows the faithfulness of the functor. And since any map between spectra $F : \Sigma^\infty X \rightarrow \Sigma^\infty Y$ restricts to the zeroth degree, which gives a map $f : X \rightarrow Y$ such that $\Sigma^\infty f = F$, the functor is also full. This completes the proof that Σ^∞ is a fully faithful functor. $\square$

A corollary that directly arises from our proof is that Σ^∞ is a left adjoint functor.

Corollary I.3.4. *Define the functor $\text{Ev}_k : \mathbf{Sp}_{\text{seq}} \rightarrow \mathbf{Top}_*$ by $\text{Ev}_k(E) = E_k$ and $\text{Ev}_k(f_\bullet) = f_k$. Then there is an adjunction of functors $\Sigma^\infty : \mathbf{Sp}_{\text{seq}} \rightleftarrows \mathbf{Top}_* : \text{Ev}_0$.*

As a sequence of pointed spaces with the structure maps, sequential spectra are natural objects to define the stable homotopy groups. In fact, the homotopy groups of a sequential spectrum are naturally stable homotopy groups if the spectra is given by the suspension of a pointed space:

Definition I.3.5 (Homotopy Groups of Spectra). For $n \in \mathbb{Z}$, one can defined the homotopy group of spectra E as

$$\pi_n(E) := \text{colim}_{k \rightarrow \infty} \pi_{n+k}(E_k),$$

where the colimit is taken for sufficiently large k such that $n + k \geq 2$, with the transition map given by

$$\pi_{n+k}(E_k) \xrightarrow{\Sigma} \pi_{n+k+1}(\Sigma E_k) \xrightarrow{(\sigma_k)_*} \pi_{n+k+1}(E_{k+1})$$

One can check that this definition is functorial, meaning that for a morphism $f : E \rightarrow F$ between spectra, there is naturally a morphism between abelian groups $\pi_n(f) : \pi_n(E) \rightarrow \pi_n(F)$, and thus, we have a functor $\pi_n : \mathbf{Sp}_{\text{seq}} \rightarrow \mathbf{Ab}$. We make the convention of notation that if we do not specify the spectra in the stable homotopy group, then we are taking the sphere spectra $\mathbb{S}$, i.e.

$$\pi_n^s := \pi_n(\mathbb{S}) = \text{colim}_{k \rightarrow \infty} \pi_{n+k}(S^k).$$

Directly following from this definition, we can make the following claim:

Proposition I.3.6. *Let X be a pointed space, $\pi_n^s(X) = \pi_n(\Sigma^\infty X)$, and thus, the Pontryagin-Thom isomorphism can be written as $\Omega_n^{\text{fr}} \cong \pi_n^s$.*

Note that in this definition of homotopy groups, we are able to define the stable homotopy group with negative degrees; this is very unusual for our intuition of spaces, and it is clear, by the following example, that the homotopy groups of spectra are not zero in general:

Example. Write $E := \Sigma^{-1}\mathbb{S} := (*, S^0, S^1, \dots)$, then

$$\begin{aligned} \pi_{-1}(E) &= \text{colim}_{k \geq 3} \pi_{k-1}(S^{k-1}) \\ &= \text{colim} \left(\mathbb{Z} \xrightarrow{\text{id}_{\mathbb{Z}}} \mathbb{Z} \xrightarrow{\text{id}_{\mathbb{Z}}} \mathbb{Z} \xrightarrow{\text{id}_{\mathbb{Z}}} \dots \right) \cong \mathbb{Z} \end{aligned}$$

which shows that $\pi_{-1}(E) = \mathbb{Z} \neq 0$.

So we defined the connectivity of the spectra:

Definition I.3.7 (Connectivity). The spectrum E is said to be connective if $\pi_n(E) = 0$ for any $n < 0$.

And it is not hard to show the following fact:

Proposition I.3.8. *For any pointed CW complex X , the suspension spectrum $\Sigma^\infty X$ is connective.*

Proof. Note that for r -connected space X , $\Sigma^k X$ is $(k+r)$ -connected. Thus, in our setup we can claim $\Sigma^k X$ is at least $(k-1)$ -connected. For the given $n < 0$ by $(k-1)$ -connectivity, there must be sufficiently large k such that $\pi_{n+k}(\Sigma^k X) = 0$, by the universal property of colimit, $\pi_n(\Sigma^\infty X) = 0$. $\square$

Note that in the category of pointed CW complexes $\mathbf{CW}_*$ (or more generally, pointed topological spaces), there is a collection of canonically identified weak equivalences $W \subseteq \text{mor}(\mathbf{CW}_*)$ ^{1.3.1} such that $f \in W$ induces an isomorphism between homotopy groups/sets:

$$f_* : \pi_k(X) \xrightarrow{\cong} \pi_k(Y), \quad \forall k \geq 0.$$

A similar definition can be made on spectra, but there are two different ways to define such equivalence:

^{1.3.1}Ignore set theoretic problems.

Definition I.3.9 (Homotopy/Level/Stable Equivalence). For sequential spectra $E, F \in \text{ob}(\mathbf{Sp}_{\text{seq}})$, one can say that the morphism $f : E \rightarrow F$ to be a

1. homotopy equivalence, if there is a homotopy inverse $g : F \rightarrow E$ such that $f \circ g \simeq \text{id}_F$ and $g \circ f \simeq \text{id}_E$ (the homotopy needs to be compatible with structure maps).
2. level equivalence, if for any $k \in \mathbb{N}$, the continuous pointed map $f_k : E_k \rightarrow F_k$ is an weak equivalence.
3. stable equivalence, if $\pi_n f : \pi_n(E) \rightarrow \pi_n(F)$

It is obvious that homotopy equivalence is the strongest definition, and stable equivalence is the weakest one. More precisely,

$$\text{homotopy equivalence} \implies \text{level equivalence} \implies \text{stable equivalence}$$

and the converse is not true in general. Here is an example:

Example. If we consider the spectrum C such that $C_k := S^k$ with the structure map be the constant map $\sigma_k^C = c_\infty : \Sigma S^k \cong S^{k+1} \rightarrow S^{k+1}$. Then, for any class $[f] \in \pi_{n+k}(C_k)$, the colimit ensures

$$\iota_k[f]_k \sim \iota_{k+1}[\sigma_k^C \circ \Sigma f]_{k+1} = [0]_{k+1}, \quad \forall [f] \in \pi_{n+k}(C_k) : \forall [\sigma_k^C \circ \Sigma f] \in \pi_{n+k+1}(C_{k+1}),$$

and thus, $\pi_n(C) = \text{colim}_k \pi_{n+k}(C_k)$. Thus, the morphism between spectra $C \rightarrow \{*\}$ is a stable equivalence, but clearly not a homotopy/level equivalence. Furthermore, note that C and $\mathbb{S}$ have exactly the same space on each level, but they are not equivalent.

Part II

Bordism Category and Field Theory

Part III

Application to Floer Theory

Part IV

Appendices

A Review of Fredholm Theory

B Thom Class and Gysin Maps

C Classifying Spaces

D The Bare Essentials of Morse and Floer Theory

This appendix is a review of basic Floer theory, which we will use without further explanation in the last part of the main text, which introduce symplectic field theory and Floer homotopy theory.

D.1 Morse-Smale-Witten Homology

Let (M, g) be a smooth Riemannian manifold with dimension n , recall that a Morse function $f \in C^\infty(M)$ on M is defined by the smooth function that all critical points are nondegenerate, i.e., for $p \in M$ such that $df_p = 0$, the Hessian can be globally defined as

$$(D.1) \quad \text{Hess}^\nabla f := \nabla(df) : TM \otimes TM \rightarrow \mathbb{R}$$

where ∇ is the Levi-Civita connection. It is a famous application of Sard's theorem in differential topology that the Morse function is generic. Note that although the Hessian clearly depends on the choice of the connection, the behavior of it on the critical point does not. More precisely, we have the following statement:

Proposition D.1. *The Hessian of $f \in C^\infty(M)$ at the critical point $p \in M$ has the following properties:*

1. *The bilinear map $\text{Hess}_p f : T_p M \times T_p M \rightarrow \mathbb{R}$ is independent of the choice of connection (and thus, not depends on the metric).*
2. *The Hessian has local coordinate representation*

$$\text{Hess}_p f(v, w) = \sum_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j} v_i w_j$$

where v_i and w_j are components of the given vectors.

Note that the second property implies the first one, and the proof is just simple calculation of local coordinates, which we omitted here. Part of the reason to study Morse function is that the Morse function has a local canonical form near critical points, which is known as the Morse lemma:

Theorem D.2 (Morse Lemma). *Let $f \in C^\infty(M)$ be a smooth function with nondegenerate critical point $p \in \text{Crit}(f)$, there exists a chart (U, φ) contains p such that f has local coordinate form*

$$f \circ \varphi^{-1}(x_1, \dots, x_n) = f(p) - x_1^2 - \dots - x_r^2 + x_{r+1}^2 + \dots + x_n^2.$$

The number $0 \leq r \leq n$ is called the Morse index of p , denote as $\text{ind}_p(f) = r$.

The proof can be done with either some complicated linear algebra technique or applying the Moser's trick, for a complete proof, see for example [1, Part I]. Other equivalent definition of Morse index includes the dimension of negative definite eigen space, and the number of negative eigenvalues of $\text{Hess}_p f$. The classical

approach to Morse theory studies the topology of a manifold through the handle attachments $D^r \times D^{n-r}$ that occur when a sublevel set crosses a critical point of index r .

After we established the local model underlying this picture the Morse lemma we can now pass directly to its homological refinement, Morse homology, using the gradient flow of Morse functions. Consider the negative gradient flow equation

$$(D.2) \quad \dot{\gamma}(s) = -\text{grad}f(\gamma(s)),$$

and let $\varphi_s : M \rightarrow M$ be the flow of Equation (D.2). Our first observation is that the critical points are hyperbolic fix points of the flow φ_s , and the energy functional is preserved for the same end points:

Proposition D.3. *With $p, q \in \text{Crit}(f)$, and the end-points condition*

$$\lim_{s \rightarrow -\infty} \gamma(s) = p, \quad \lim_{s \rightarrow \infty} \gamma(s) = q.$$

Then the energy functional of the flow line is preserved, and explicitly given by

$$E(\gamma) := \int_{\mathbb{R}} \|\dot{\gamma}(s)\|^2 ds = f(p) - f(q)$$

Proof. The proof is just simple computation, note that

$$\begin{aligned} \frac{d}{ds} f(\gamma(s)) &= df_{\gamma(s)}(\dot{\gamma}(s)) = \langle \text{grad}_{\gamma(s)} f, \dot{\gamma}(s) \rangle \\ &= \langle -\dot{\gamma}(s), \dot{\gamma}(s) \rangle = -\|\dot{\gamma}(s)\|^2. \end{aligned}$$

Then, the energy functional is given by

$$E(\gamma) = \int_{\mathbb{R}} \|\dot{\gamma}(s)\|^2 ds = - \int_{\mathbb{R}} \frac{d}{ds} f(\gamma(s)) ds = f(p) - f(q).$$

This completes the proof. □

This is already a finite dimensional prototype of Floer energy identity. It leads to the following properties

1. f strictly decrease along non-constant trajectory.
2. The flow line only flows from larger critical values to smaller critical values.
3. If f takes the same value on both end points, i.e., $f(p) = f(q)$, then the trajectory is constant.

Using the gradient flow equation Equation (D.2), we may define the stable and unstable manifolds, which collect all fixed point (critical points) information of the gradient flow:

Definition D.4. Let $f \in C^\infty(M)$, the stable and unstable manifolds for critical point $p \in \text{Crit}(f)$ are defined by the following submanifolds of M

$$\begin{aligned} W^s(p; f) &:= \left\{ x \in M \mid \lim_{s \rightarrow \infty} \varphi_s(x) = p \right\} \\ W^u(q; f) &:= \left\{ y \in M \mid \lim_{s \rightarrow -\infty} \varphi_s(y) = q \right\} \end{aligned}$$

Note that they are both smooth injectively immersed submanifold of M . More precisely,

Proposition D.5. *Let $f \in C^\infty(M)$ be a Morse function with critical point $p \in \text{Crit}(f)$. Then, the dimension of stable and unstable manifold is given by*

$$\dim W^u(p; f) = \text{ind}_p(f), \quad \dim W^s(p; f) = n - \text{ind}_p(f).$$

The proof in general cases is rather complicated, however, this fact is intuitive in the following sense: Given an open neighborhood U of the critical point p with a Euclidean metric, by Morse Lemma [Theorem D.2](#), the Morse function under some local coordinate is given by

$$f(x, y) = f(p) + \frac{1}{2}\|x\|^2 + \frac{1}{2}\|y\|^2.$$

Under this choice, the gradient of f is simply $\text{grad} f = (-x, y)$, and the gradient flow equation has solution

$$(x(t), y(t)) = (e^t x_0, e^{-t} x_0).$$

With this solution, the statement is straightforward at least locally:

1. As $t \rightarrow \infty$, the trajectory $(x, y) \rightarrow (0, 0)$ if and only if $x_0 = 0$. This gives the local stable manifold $W_{\text{loc}}^s(p; f) = \{(0, y) \mid y \in \mathbb{R}^{n-\lambda}\}$.
2. As $t \rightarrow -\infty$, the trajectory $(x, y) \rightarrow (0, 0)$ if and only if $y_0 = 0$. This gives the local unstable manifold $W_{\text{loc}}^u(p; f) = \{(x, 0) \mid x \in \mathbb{R}^\lambda\}$.

And one can get the dimension of the stable and unstable manifolds simply by the definition of local manifolds above, and note that this also shows $T_p W^s(p; f) \cong E_p^+$ and $T_p W^u(p; f) \cong E_p^-$, where $T_p M \cong E_p^+ \oplus E_p^-$ is the splitting given by the positive and negative eigenspaces of $\text{Hess}_p(f)$. The fact that the flow φ_s is local diffeomorphism allow us to claim

$$W^s(p; f) = \bigcup_{t \geq 0} \varphi_{-t}(W_{\text{loc}}^s(g; f)), \quad W^u(p; f) = \bigcup_{t \geq 0} \varphi_t(W_{\text{loc}}^u(g; f))$$

One can try to establish a metric invariant statement to make the method above be a general claim, and we may omit any further detail of the proof here.

Definition D.6 (Morse-Smale Pair). A pair (f, g) of smooth function f and Riemannian metric g is said to be satisfies the Morse-Smale condition if

$$W^s(p; f) \pitchfork W^u(q; f), \quad p, q \in \text{Crit}(f).$$

Such a pair is called a Morse-Smale pair.

With the transversal intersection condition, we can define the parametrized space of the gradient flow trajectory:

Definition D.7 (Parameterized Trajectory Space). The parameterized trajectory space of the gradient flow between $p, q \in \text{Crit}(f)$ is defined as

$$\widehat{\mathcal{M}}(p, q; f) := W^s(q; f) \cap W^s(p; f).$$

With some effort, we can show that this space is indeed the following submanifold: Let $W_f(p, q)$ denote the set of trajectories from p to q along $-\nabla f$

$$(D.3) \quad \widehat{\mathcal{M}}(p, q; f) = \{x \in M \mid \exists \gamma \in W_f(p, q) : \exists t \in (-\epsilon, \epsilon) : \gamma(t) = x\},$$

and its dimension is given by $\dim \widehat{\mathcal{M}}(p, q; f) = \text{ind}_p(f) - \text{ind}_q(f)$. The identification in [Equation \(D.3\)](#) is due to the following proposition:

Proposition D.8. *Let M be a compact manifold and (f, g) be a Morse-Smale pair. Then for any negative gradient trajectory $\gamma : \mathbb{R} \rightarrow M$ such that $\dot{\gamma} = -\nabla f \circ \gamma$*

Also, a trajectory has a freedom of translation, where $\mathbb{R} \curvearrowright \widehat{\mathcal{M}}(p, q; f)$ acts freely by

$$t \cdot \gamma(s) := \gamma(t + s).$$

What we want to study is really the orbit of this $\mathbb{R}$ action:

Definition D.9 (Morse Moduli Space). Defined the unparametrized Morse moduli space of Morse-Smale pair (f, g) and $p, q \in \text{Crit}(f)$ to be $\mathcal{M}(p, q; f) := \widehat{\mathcal{M}}(p, q; f)/\mathbb{R}$.

An intuitive fact is that the dimension of this moduli space is

$$\dim \mathcal{M}(p, q; f) = \text{ind}_p(f) - \text{ind}_q(f) - 1.$$

One can prove this fact by considering the short exact sequence of tangent space induced by the inclusion of orbits.

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